

Project Proposal: semantic segmentation of clouds in IR images

Objects that create obscurations in aerial/satellite images interfere with computer vision algorithms, and may lead to poor results in automatic surface analyses. We would like to address the issue of occlusions caused by clouds, and semantic segmentate them.

Our goal is to use fine-tuning techniques in order to improve the performance of a pre-trained model over datasets of different domain: low-altitude airborne IR images. We will use a private annotated dataset from our workplace (about 5000 images).

We chose to use a Unet based model according to the work presented in the following article [1]. The dataset of this work consists of satellite images ("38-cloud dataset"). The model is implemented with tensorflow, including published weights.

We considered two more (newer) articles and architectures ([2], [3]), but due to lack of published implementations and weights we selected this one.

We will use Intersection over Union (IoU) or Dice scores in order to evaluate the model (in addition to the classic metrics of precision, recall, etc.).

Gil Grimberg, gilgrimberg@mail.tau.ac.il, 204293781

Tifferet Nevo, tifferetnevo@mail.tau.ac.il, 314735572

[1] CLOUD-NET: AN END-TO-END CLOUD DETECTION ALGORITHM FOR
LANDSAT 8 IMAGERY

<https://arxiv.org/pdf/1901.10077>

[2]

Cloud segmentation in satellite imagery using attention-based deep learning, 2025:

<https://link.springer.com/article/10.1007/s11042-025-20860-9>

[3]

Semantic segmentation of clouds in satellite images based on U-Net++ architecture and attention mechanism, 2022:

<https://www.sciencedirect.com/science/article/abs/pii/S0957417422014932#previewsection-references>